

Hampton University R and Biostatistics Workshop

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Overview

1. **Introductions**
2. **Quick pre-workshop survey**
3. **Workshop schedule**
4. **Workshop materials**
5. **Post workshop survey**

Introductions

Name

Background

Research Interests

Pre-workshop survey

Please take a few moments to fill out this pre-workshop survey.

Survey Link: <https://redcap.link/nf9crpm6>



Workshop Schedule and Material

1. Day 1: Instruction
2. Day 2: Implementation and special topics
3. Follow along: <https://sl-itw.github.io/HU-Workshop/>
4. Material:

Download R and
Rstudio: <https://posit.co/download/rstudio-desktop/>

AirBNB data
set: <https://data.insideairbnb.com/united-states/ny/new-york-city/2024-07-05/visualisations/listings.csv>

Census API
key: https://api.census.gov/data/key_signup.html

Pharmacy
Data: <https://archive.icpsr.umich.edu/nanda/view/studies/120907>

Date	Time	Topic
DAY 1 Tuesday, July 30, 2024	WORKSHOP I: 9:00 am-11:30 am	Data Sourcing and Wrangling
	11:30 am-12:45 pm	LUNCH
	WORKSHOP II: 1:00 pm-3:30 pm	Exploratory Data Analysis
DAY 2 Wednesday, July 31, 2024 **Research Consultations may be Individual and/or Groups**	WORKSHOP III: 9:00 am-11:00 am	Interactive Team Science-based real-world application of data wrangling and exploratory data analysis techniques learned from Day 1
	11:00 pm-12:45 pm	Lunch & Research Consultations
	Workshop IV: 1:00 pm-3:00 pm	Advanced topics in statistical analysis & programming informed by relevant interests of workshop participants
	3:00 pm-4:00 pm	Research Consultations
	4:00 pm-5:00 pm	<i>Data Analytics consultation with Academic Affairs, Student Affairs & Assessment</i>

What is Biostatistics

Biostatistics is the science of studying phenomena in biological data

- It brings context to the math associated by providing tools to measure uncertainty
- Ultimately biostatisticians tend to collaborate with other disciplines to solve biological problems and play key roles in study design, and evaluation
- R is a very popular coding language for doing such work

Open-Source Statistical Language - R

R is mainly used as an object-oriented statistical coding language. This means that we read-in or create objects and then perform functions on them.

- Users create and upload packages to CRAN which manages the R platform
- Packages are a set of functions that are supposed to work well together to complete a task
- At its core, R is a very sophisticated calculator

Morning Session

Data sourcing and wrangling

Afternoon Session

Exploratory Data Analysis

Morning Session

**Geospatially
modeling pharmacy access adjusting
for other census tract level social variables**

Data Sources and Study Variables

Data source (NaNDA, ACS)

- Virginia census tracts (study population)
 - A census tract is a small, relatively permanent statistical subdivision of a county or similar entity in Virginia
- Pharmacy count (outcome)
 - Number of pharmacies in a census tract
- Population (offset)
- Race ethnicity percentages
- Mobility
- Social Economic Status variables
- Age

Areal data and irregular lattices

- **Areal Data: Data associated with geographical areas**
 - Example: Number of pharmacies in different neighborhoods
- **Irregular lattices are spatial structures where areas are of different shapes and sizes.**
 - Example: Neighborhoods in a city, where each neighborhood may have a different shape and size
- **Modeling areal data assumptions:**
 - Spatial dependency: Areas close to each other are likely to have similar values
 - We will use spatial random effects to capture these dependencies

A note on why bayesian vs frequentist approach

1. **Data Generating Function:** Likelihood function that describes how the data is generated from the parameters.

Example: Poisson distribution for count data like the number of pharmacies.

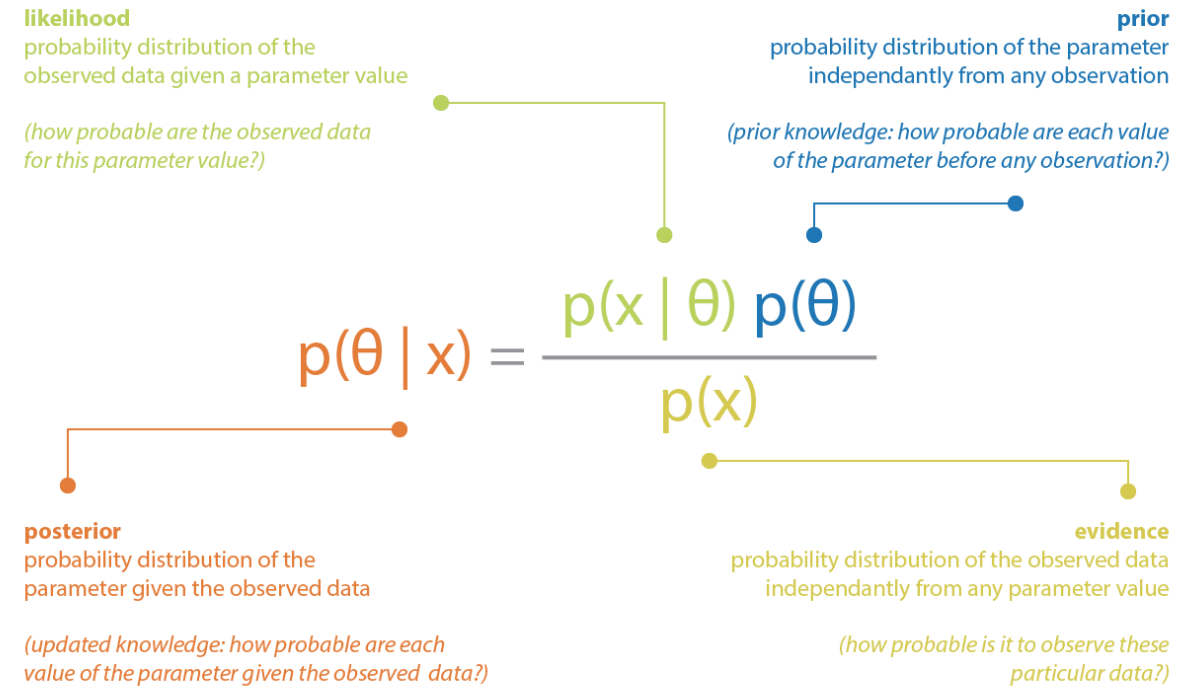
2. **Priors:** Incorporate prior knowledge or assumptions about the parameters.

Example: Spatial priors can capture the spatial dependencies between areas

3. **Credible vs confidence intervals:** Both are bounds of attributed uncertainty or risk

Example: Credible intervals represent 95% probability of the estimate existing in a bound while confidence intervals assumes if a test is repeated multiple times it will be within the bound 95% of the time

4. **Frequentist approach favors asymptotic results, while the Bayesian approach favors data-driven inference incorporating prior knowledge.**



Morning Session

Lets head over to R Studio

What spatial modeling fails to explain

Vulnerable populations may be overlooked because modeling tends to focus on averages over the whole study area and not able to identify the unique challenges of some low access neighborhoods compared to others

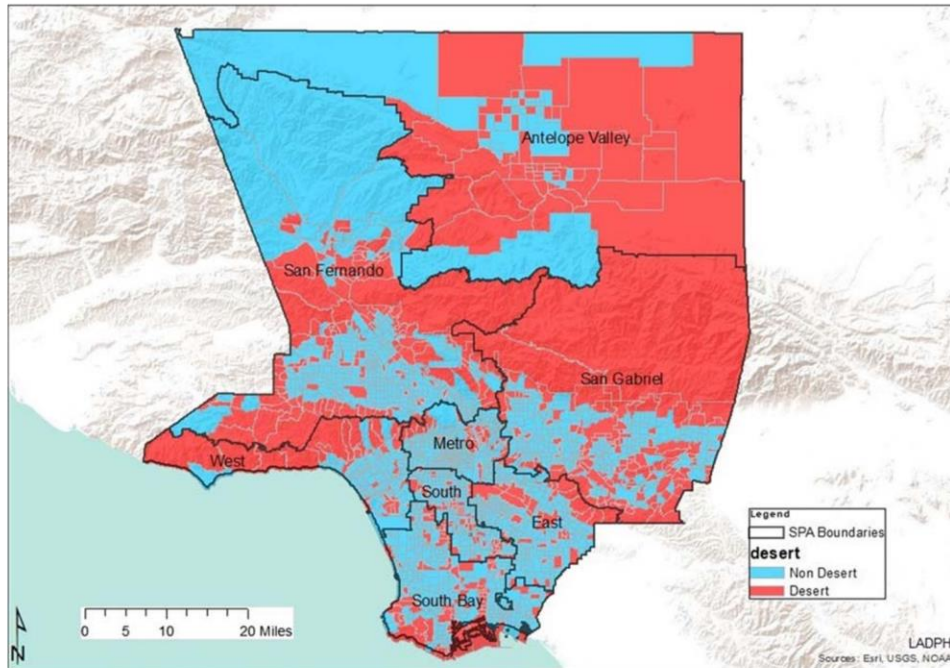
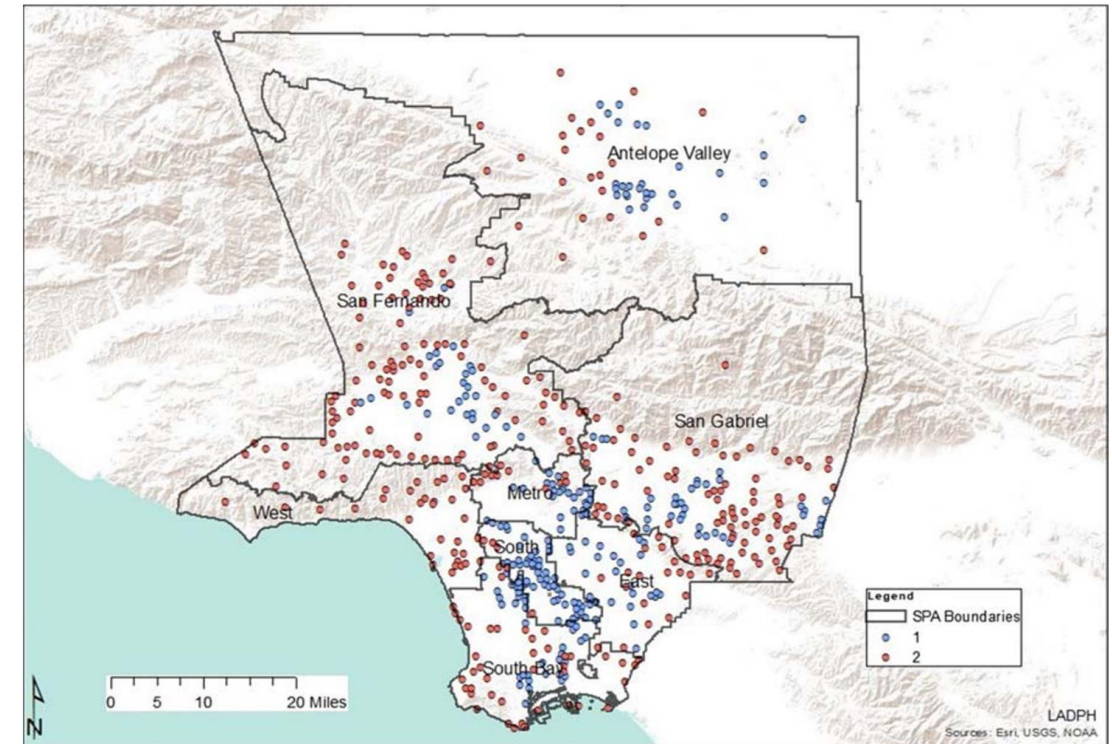


Fig. 1 Los Angeles County pharmacy deserts and pharmacy non-deserts by Service Planning Area



Pharmacy desert types by Service Planning Area

Wisseh, C., Hildreth, K., Marshall, J., Tanner, A., Bazargan, M., & Robinson, P. (2021). Social Determinants of Pharmacy Deserts in Los Angeles County. *Journal of Racial and Ethnic Health Disparities*, 8(6), 1424–1434. <https://doi.org/10.1007/s40615-020-00904-6>

What spatial modeling fails to explain

Type 2 deserts tend to have a smaller population size, mostly elderly, and low poverty. The authors suggest type two deserts would benefit from proximity-based interventions.

Type 1 deserts tend to have more compounding issues and may require a lot more interventions on multiple fronts.

Table 3 Social determinants of health elements by pharmacy desert type

			Type 1 desert (<i>N</i> = 238)		Type 2 desert (<i>N</i> = 333)	
			Residents (<i>n</i>)	<i>R</i> ²	Residents (<i>n</i>)	Share
Healthy People 2020 SDOH Framework	Social and community context	Population density	7753*	0.99	2806	0.65
		Hispanic	737,380	0.82	362,410	0.94
		Non-Hispanic Black	126,402	0.92	78,749	1.00
		Younger than 18 years of age	289,409	0.54	285,036	1.00
		Older than 64 years of age	100,522	1.00	227,329	1.00
	Economic stability	Below federal poverty level	237,266	0.71	103,792	1.00
		Rent and do not own home	549,198	0.99	341,942	0.96
		Education				
	Neighborhood and environment	No high school diploma	224,904	0.86	81,298	0.63
		English as a second language	652,985	0.83	500,066	0.85
		No vehicle	26,884	1.00	15,143	0.38
	Health care	Crimes against property	7.10**	0.67	4.32	0.89
		Crimes against people	6.74**	0.78	3.67	0.89
		Health professional shortage area	0.76**	1.00	0.10	1.00
		No health insurance	143,906	1.00	71,230	0.74

*Residents per square mile

**Mean index

Wisseh, C., Hildreth, K., Marshall, J., Tanner, A., Bazargan, M., & Robinson, P. (2021). Social Determinants of Pharmacy Deserts in Los Angeles County. *Journal of Racial and Ethnic Health Disparities*, 8(6), 1424–1434. <https://doi.org/10.1007/s40615-020-00904-6>

Assessing pharmacy access and health outcomes

- Primary outcome: medication adherence
- Primary exposure: pharmacy closure
- Conclusion: "The mean association of pharmacy closure with adherence was greater among individuals using independent pharmacies (-7.89% ; 95% CI, -8.32% to -7.47%) or living in neighborhoods with fewer pharmacies (-7.98% ; 95% CI, -8.50% to -7.47%) compared with their counterparts"

Qato, D. M., Alexander, G. C., Chakraborty, A., Guadamuz, J. S., & Jackson, J. W. (2019). Association between pharmacy closures and adherence to cardiovascular medications among older US adults. *JAMA network open*, 2(4), e192606-e192606.

Thank you

